Regularized different-sized RPY pair mobilities in the Kim & Karrila irreducible basis

Translation–translation, translation–rotation, and translation–dipole couplings

Purpose and conventions

These notes transcribe the different-sized regularized RPY mobility tensors of Zuk, Wajnryb, Mizerski & Szymczak (*J. Fluid Mech.* **741**, R5, 2014) into the Kim & Karrila (K&K) scalar mobility functions, suitable for a Stokesian-dynamics code that stores pair mobilities in the irreducible (x, y) form. For each pair (i, j) we write $\mathbf{R}_{ij} = \mathbf{R}_i - \mathbf{R}_j$, $R = |\mathbf{R}_{ij}|$, $\hat{\mathbf{R}} = \mathbf{R}_{ij}/R$; a_i, a_j are the two radii and $a_>, a_<$ the larger/smaller of the pair. All expressions are the open-boundary (free-space) propagator forms; η is the solvent viscosity, and Θ is the Heaviside step function.

Each coupling has three branches: the non-overlapping far-field form ($R > a_i + a_j$), the overlapping regularized form ($a_> - a_< < R \le a_i + a_j$), and the complete-overlap form ($R \le a_> - a_<$, one sphere inside the other).

Basis tensors. The three couplings are decomposed as

$$\boldsymbol{\mu}^{tt}_{ij} = x^a_{ij}\,\hat{\mathbf{R}}\hat{\mathbf{R}} + y^a_{ij}\,(\mathbf{1} - \hat{\mathbf{R}}\hat{\mathbf{R}}), \tag{1}$$

$$\boldsymbol{\mu}^{rt}_{ij} = y^b_{ij}\,(\boldsymbol{\epsilon}\cdot\hat{\mathbf{R}}), \qquad [\boldsymbol{\epsilon}\cdot\hat{\mathbf{R}}]_{\alpha\beta} = \epsilon_{\alpha\beta\gamma}\hat{R}_\gamma, \tag{2}$$

$$\mu^{td}_{ij,\alpha\beta\gamma} = x^g_{ij}\Big(\hat{R}_\beta\hat{R}_\gamma - \tfrac{1}{3}\delta_{\beta\gamma}\Big)\hat{R}_\alpha + y^g_{ij}\Big(\delta_{\alpha\beta}\hat{R}_\gamma + \delta_{\alpha\gamma}\hat{R}_\beta - 2\hat{R}_\alpha\hat{R}_\beta\hat{R}_\gamma\Big). \tag{3}$$

In (3) the first index α is the translational (velocity) index and the pair (β, γ) contracts with the symmetric traceless rate-of-strain $E_{\beta\gamma}$ (equivalently the stresslet). Both basis tensors are symmetric and traceless in (β, γ). This is the standard K&K $gt/\tilde{g}$ decomposition; see the remark at the end regarding the normalization of the y^g basis tensor.

1 Translation–translation coupling $\boldsymbol{\mu}^{tt}$

Zuk et al. (3.1), in the $\{\mathbf{1}, \hat{\mathbf{R}}\hat{\mathbf{R}}\}$ form:

$$\boldsymbol{\mu}^{tt}_{ij} = \begin{cases} \dfrac{1}{8\pi\eta R}\left[\left(1 + \dfrac{a_i^2 + a_j^2}{3R^2}\right)\mathbf{1} + \left(1 - \dfrac{a_i^2 + a_j^2}{R^2}\right)\hat{\mathbf{R}}\hat{\mathbf{R}}\right], & a_i + a_j < R, \\[2ex] \dfrac{1}{6\pi\eta a_i a_j}\left[\begin{array}{l} \frac{16R^3(a_i+a_j) - \left((a_i-a_j)^2 + 3R^2\right)^2}{32R^3}\,\mathbf{1} \\ + \frac{3\left((a_i-a_j)^2 - R^2\right)^2}{32R^3}\,\hat{\mathbf{R}}\hat{\mathbf{R}} \end{array}\right], & a_> - a_< < R \le a_i + a_j, \\[2ex] \dfrac{1}{6\pi\eta a_>}\,\mathbf{1}, & R \le a_> - a_<. \end{cases} \tag{4}$$

Writing $\boldsymbol{\mu}^{tt} = x^a\hat{\mathbf{R}}\hat{\mathbf{R}} + y^a(\mathbf{1} - \hat{\mathbf{R}}\hat{\mathbf{R}})$ (so x^a is the longitudinal / $\hat{\mathbf{R}}\hat{\mathbf{R}}$ eigenvalue and y^a the transverse one), one has y^a = (coeff of $\mathbf{1}$) and x^a = (coeff of $\mathbf{1}$) + (coeff of $\hat{\mathbf{R}}\hat{\mathbf{R}}$). For the non-overlapping branch ($a_i + a_j < R$):

$$\boxed{x^a_{ij} = \frac{1}{4\pi\eta R}\left(1 - \frac{a_i^2 + a_j^2}{3R^2}\right), \qquad y^a_{ij} = \frac{1}{8\pi\eta R}\left(1 + \frac{a_i^2 + a_j^2}{3R^2}\right).} \tag{5}$$

For the overlapping branch ($a_> - a_< < R \le a_i + a_j$):

$$\boxed{\begin{aligned} x^a_{ij} &= \frac{16R^3(a_i + a_j) + 2(a_i - a_j)^4 - 12(a_i - a_j)^2 R^2 - 6R^4}{192\pi\eta\, a_i a_j\, R^3}, \\ y^a_{ij} &= \frac{16R^3(a_i + a_j) - \big((a_i - a_j)^2 + 3R^2\big)^2}{192\pi\eta\, a_i a_j\, R^3}. \end{aligned}} \tag{6}$$

For the complete-overlap branch ($R \le a_> - a_<$): $x^a_{ij} = y^a_{ij} = 1/(6\pi\eta a_>)$.

2 Translation–rotation coupling $\boldsymbol{\mu}^{rt}$

Zuk et al. (3.5) gives $\boldsymbol{\mu}^{rt}_{ij} = y^b_{ij}\,(\boldsymbol{\epsilon}\cdot\hat{\mathbf{R}})$ with

$$\boxed{y^b_{ij} = \begin{cases} \dfrac{1}{8\pi\eta R^2}, & a_i + a_j < R, \\ \dfrac{(a_i - a_j + R)^2\big(a_j^2 + 2a_j(a_i + R) - 3(a_i - R)^2\big)}{128\pi\eta\, a_i^3 a_j\, R^2}, & a_> - a_< < R \le a_i + a_j, \\ \Theta(a_i - a_j)\,\dfrac{R}{8\pi\eta a_i^3}, & R \le a_> - a_<. \end{cases}} \tag{7}$$

The complete-overlap branch uses $\zeta^{rr}_i = 8\pi\eta a_i^3$. The translation–rotation partner $\boldsymbol{\mu}^{tr}_{ij}$ has the identical form with a_i and a_j interchanged (Zuk et al. note that $\boldsymbol{\mu}^{tr}$ is *not* simply the transpose of $\boldsymbol{\mu}^{rt}$, correcting a misprint in Wajnryb et al. 2013).

3 Translation–dipole coupling $\boldsymbol{\mu}^{td} \to (x^g, y^g)$

The mapping

Zuk et al. (3.6) write the coupling in the non-irreducible form

$$\mu^{td}_{ij,\alpha\beta\gamma} = C_1\,\delta_{\alpha\beta}\hat{R}_\gamma + C_2\,\hat{R}_\alpha\hat{R}_\beta\hat{R}_\gamma, \tag{8}$$

where $[\mathbf{1}\hat{\mathbf{R}}]_{\alpha\beta\gamma} = \delta_{\alpha\beta}\hat{R}_\gamma$ in their notation. Because μ^{td} is contracted with the symmetric traceless $E_{\beta\gamma}$, only the part of (8) that is symmetric and traceless in (β,γ) is physical; Zuk et al. deliberately record the simplest algebraic representative. Contracting both (8) and the basis form (3) with an arbitrary symmetric traceless E gives

$$[\mu^{td}\!:\!E]_\alpha = \underbrace{C_2}_{\text{Zuk}}\,\hat{R}_\alpha(\hat{\mathbf{R}}\hat{\mathbf{R}}\!:\!E) + \underbrace{C_1}_{\text{Zuk}}\,\hat{R}_\gamma E_{\alpha\gamma} = \underbrace{(x^g - 2y^g)}_{\text{K\&K}}\,\hat{R}_\alpha(\hat{\mathbf{R}}\hat{\mathbf{R}}\!:\!E) + \underbrace{2y^g}_{\text{K\&K}}\,\hat{R}_\gamma E_{\alpha\gamma}. \tag{9}$$

Matching the two independent structures $\hat{R}_\alpha(\hat{\mathbf{R}}\hat{\mathbf{R}}\!:\!E)$ and $\hat{R}_\gamma E_{\alpha\gamma}$ yields the linear relation

$$\boxed{x^g_{ij} = C_1 + C_2, \qquad y^g_{ij} = \tfrac{1}{2}\,C_1\,.} \tag{10}$$

This was verified numerically to machine precision against the full Cartesian tensor (8) in every branch, and its far-field limit reproduces the expected $x^g \to -5a_j^3/(2R^2)$, $y^g = O(R^{-4})$.

Branch 1 — non-overlapping, $R > a_i + a_j$

Here $C_1 = -\dfrac{a_j^3(5a_i^2 + 3a_j^2)}{3R^4}$ and $C_2 = \dfrac{5a_j^3(5a_i^2 + 3a_j^2 - 3R^2)}{6R^4}$, so (10) gives

$$\boxed{x^g_{ij} = \frac{a_j^3\big(5a_i^2 + 3a_j^2 - 5R^2\big)}{2R^4}, \qquad y^g_{ij} = -\frac{a_j^3\big(5a_i^2 + 3a_j^2\big)}{6R^4}} \qquad (R > a_i + a_j). \tag{11}$$

Branch 2 — overlapping, $a_> - a_< < R \le a_i + a_j$

Zuk et al. write $\boldsymbol{\mu}^{td}_{ij} = \frac{5}{6}a_j\big[\mathscr{C}\,\mathbf{1}\hat{\mathbf{R}} + \mathscr{D}\,\hat{\mathbf{R}}\hat{\mathbf{R}}\hat{\mathbf{R}}\big]$, i.e. $C_1 = \frac{5}{6}a_j\mathscr{C}$, $C_2 = \frac{5}{6}a_j\mathscr{D}$, with

$$\mathscr{C} = \frac{10R^6 - 24a_iR^5 - 15R^4(a_j^2 - a_i^2) + (a_j - a_i)^5(a_i + 5a_j)}{40\,a_ia_jR^4}, \tag{12}$$

$$\mathscr{D} = \frac{\big((a_i - a_j)^2 - R^2\big)^2\big((a_i - a_j)(a_i + 5a_j) - R^2\big)}{16\,a_ia_jR^4}. \tag{13}$$

Then (10) gives, compactly,

$$\boxed{x^g_{ij} = \frac{5}{6}\,a_j\big(\mathscr{C} + \mathscr{D}\big), \qquad y^g_{ij} = \frac{5}{12}\,a_j\,\mathscr{C}} \qquad (a_> - a_< < R \le a_i + a_j). \tag{14}$$

Explicitly, in terms of a_i, a_j, R (writing $P \equiv 10R^6 - 24a_iR^5 - 15R^4(a_j^2 - a_i^2) + (a_j - a_i)^5(a_i + 5a_j)$ and $Q \equiv \big((a_i - a_j)^2 - R^2\big)^2\big((a_i - a_j)(a_i + 5a_j) - R^2\big)$),

$$y^g_{ij} = \frac{P}{96\,a_iR^4}, \tag{15}$$

$$x^g_{ij} = 2\,y^g_{ij} + \frac{5a_j}{6}\,\mathscr{D} = \frac{P}{48\,a_iR^4} + \frac{5\,Q}{96\,a_iR^4} = \frac{2P + 5Q}{96\,a_iR^4}. \tag{16}$$

Branch 3 — complete overlap, $R \le a_> - a_<$

Zuk et al. give $\boldsymbol{\mu}^{td}_{ij} = -\Theta(a_j - a_i)\,R\,\mathbf{1}\hat{\mathbf{R}}$, i.e. $C_1 = -\Theta(a_j - a_i)\,R$, $C_2 = 0$, so

$$\boxed{x^g_{ij} = -\Theta(a_j - a_i)\,R, \qquad y^g_{ij} = -\tfrac{1}{2}\,\Theta(a_j - a_i)\,R} \qquad (R \le a_> - a_<). \tag{17}$$

This is non-zero only when the dipole-bearing sphere j is the larger one (so that sphere i is immersed in it); otherwise the coupling vanishes. All branches are continuous at $R = a_i + a_j$ and $R = a_> - a_<$ (verified numerically).

Remarks for implementation

- **Index/particle roles.** In $\boldsymbol{\mu}^{td}_{ij}$, particle i carries the translational velocity (Faxén response) and particle j carries the dipole/stresslet; the asymmetry $a_i \leftrightarrow a_j$ is genuine, so populate the transpose block ($\boldsymbol{\mu}^{dt}$, strain on i from force on j) with the roles swapped rather than by naive transposition.

- y^g **basis normalization.** The relations (10) assume the y^g basis tensor $\big(\delta_{\alpha\beta}\hat{R}_\gamma + \delta_{\alpha\gamma}\hat{R}_\beta - 2\hat{R}_\alpha\hat{R}_\beta\hat{R}_\gamma\big)$ exactly as in (3). If your code instead defines this basis tensor with a factor $\frac{1}{2}$ (i.e. $\frac{1}{2}(\delta_{\alpha\beta}\hat{R}_\gamma + \delta_{\alpha\gamma}\hat{R}_\beta) - \hat{R}_\alpha\hat{R}_\beta\hat{R}_\gamma$), multiply the tabulated y^g by 2. The x^g basis tensor $(\hat{R}_\beta\hat{R}_\gamma - \frac{1}{3}\delta_{\beta\gamma})\hat{R}_\alpha$ is unambiguous.

- **Consistency with the tt substitution rule.** For the non-overlapping $\boldsymbol{\mu}^{tt}$ only, the different-sized result follows from the equal-sized one via $a^2 \to (a_i^2 + a_j^2)/2$; this shortcut fails for all overlapping branches and for the rt and td couplings, which is why the explicit forms above are needed.